California House Price Prediction Using Machine Learning

B.Tech Computer Science & Engineering — Machine Learning Project Report
Course Outcome (CO5): Solve real-life problems using appropriate machine learning models and evaluate performance measures.

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1. Executive Summary & Problem Statement

Accurate real estate valuation is a fundamental problem in computational economics and predictive modeling. This project develops an end-to-end Machine Learning pipeline and RESTful API to predict median house prices in California census districts based on demographic, socioeconomic, and geographic factors. The system compares multiple regression algorithms, evaluates them using 5-Fold Cross-Validation and standard metrics (MAE, RMSE, R^2), serializes the optimal model (model.pkl), and deploys it as a scalable Flask service.

2. Dataset Overview (California Housing)

The dataset is derived from the 1990 U.S. Census via Scikit-Learn (fetch_california_housing). It consists of **20,640 records** with 8 continuous features and a target variable (MedHouseVal) denominated in units of $100,000 USD.

Feature Name	Description	Unit / Scale	Range
MedInc	Median Income in block group	$10,000s	0.5 – 15.0
HouseAge	Median house age	Years	1 – 52
AveRooms	Average rooms per household	Count	2.0 – 14.0
AveBedrms	Average bedrooms per household	Count	0.5 – 6.0
Population	Block group total population	Headcount	3 – 35,000
AveOccup	Average household occupancy	Members	1.0 – 5,000
Latitude	Block group centroid latitude	Degrees (N)	32.5 – 42.0
Longitude	Block group centroid longitude	Degrees (W)	-124.0 – -114.0

3. Algorithm Evaluation & Cross-Validation Results

Data was partitioned using an **80/20 train/test split**. Three regression pipelines with median imputation and feature scaling were evaluated using **5-Fold Cross-Validation** on the training set to prevent data leakage.

Model Name	5-Fold CV RMSE	Test MAE ($100k)	Test RMSE ($100k)	Test R^2 Score	Selection
Linear Regression	0.7205 (±0.0139)	0.5332 ($53.3k)	0.7455 ($74.5k)	0.5757	Baseline
Random Forest Regressor	0.5141 (±0.0114)	0.3340 ($33.4k)	0.5052 ($50.5k)	0.8051	Strong Candidate
Gradient Boosting Regressor	0.4808 (±0.0127)	0.3209 ($32.1k)	0.4808 ($48.1k)	0.8236	■ Selected Best

4. REST API Specification & Deployment

The serialized model (model.pkl) is served via app.py on Flask. It exposes GET /health and POST /predict endpoints with full range and type validation.

Sample Request (JSON):

```
{"MedInc": 6.2, "HouseAge": 20.0, "AveRooms": 6.5, "AveBedrms": 3.0, "Population": 1200.0,
"AveOccup": 3.4, "Latitude": 34.2, "Longitude": -118.4}
```

Sample Response (JSON):

```
{"success": true, "predicted_value": 2.7845, "predicted_price_usd": 278451.17,
"formatted_price": "$278,451.17", "model_name": "Gradient Boosting Regressor"}
```


5. Conclusion & Academic Attainment

The project successfully attains **Course Outcome CO5** by applying machine learning to real-world housing data, scientifically comparing candidate models, optimizing hyperparameters, and deploying an integrated REST service.